Ian Drosos

[LinkedIn](#) [Google Scholar](#)

UX design engineering lead, front-end architect, and 0-to-1 product engineer specializing in human-centered AI. I combine product design, mixed-methods HCI research, and hands-on engineering to make complex agentic systems understandable, steerable, and useful.

01 / EXPERIENCE

Trent AI, London, UK [7/2025-PRESENT]

Member of Technical Staff

- Lead UX design engineering as the primary designer and lead front-end engineer, taking Trent's core agentic security product from 0 to 1.
- Own the React/TypeScript application architecture, reusable component system, and integrations with rapidly evolving agent capabilities.
- Design and ship generative UX patterns, including chat-driven interface adaptation, human approval for agent-proposed actions, and interactive security plans, goals, and assessments.
- Drive product redesigns that reduce jargon, alert noise, and cognitive load while preserving the context users need for high-stakes security decisions.

Microsoft Research, Cambridge, UK [7/2022-7/2025]

Researcher

- Designed, implemented, and evaluated (n=16) a React-based LLM system that generated contextual prompt controls, later released as Promptions, Microsoft's open-source framework for dynamic conversational AI interfaces.
- Led UX research (n=24) on an LLM prototype that generated provocations to support critical thinking during data-driven decision-making.
- Co-authored 12 HCI publications, co-invented four patents, and translated research findings for Microsoft product leadership.

Autodesk Research, Remote, US [1/2021-4/2021]

User Interface Research Intern

- Investigated barriers to expert support in Autodesk Fusion 360 and designed and deployed a custom survey prototype with experts (n=28); results published at NordiCHI 2026.

Microsoft, Redmond, WA [7/2018-12/2018]

Research Intern

- Designed, implemented, and evaluated Wrex (n=12), a programming-by-example system for generating readable Python in Jupyter notebooks; participants completed data-wrangling tasks more effectively, and the resulting CHI 2020 paper received a Best Paper Award.

UCSD - The Design Lab, La Jolla, CA [9/2017-6/2022]

PhD Researcher

- Conducted HCI research on transparent, inspectable workflows for learning and doing data science, resulting in six publications and two paper awards.
- Served as instructor of record for HCI Portfolio Design Studio and supported courses in interaction design, HCI programming, and data-driven UX/product design.

Verizon, Alpharetta, GA [5/2011-7/2015]

Member Technical Staff I & II, System Engineering

- Built and maintained full-stack Java, JavaScript, and PL/SQL systems supporting enterprise accounts, contracts, and purchase orders across the software-development lifecycle.
- Modernized legacy applications and resolved critical production issues, including stabilizing systems during a major product rollout.

02 / SKILLS

Product & UX: UX leadership; 0-to-1 product design; interaction design; prototyping; workflow simplification; Figma

AI & Security UX: agentic AI; generative and dynamic interfaces; conversational UX; AI steering; human-in-the-loop controls; high-stakes decision support

Front-End Architecture: React; TypeScript; JavaScript; application architecture; reusable component systems; interaction architecture; API integration

Research: mixed-methods HCI; study design; interviews; surveys; experiments; usability testing; thematic and content analysis; comparative system evaluation

Additional Engineering and Analysis: Python; Java; SQL; R

03 / EDUCATION

University of California San Diego, La Jolla, CA [2017-2022]

PhD, Cognitive Science

Thesis: Synthesizing Transparent and Inspectable Technical Workflows

North Carolina State University, Raleigh, NC [2015-2017]

MS, Computer Science

Thesis: HappyFace: Identifying and Predicting Frustrating Learning Obstacles at Scale

Southern Polytechnic State University*, Marietta, GA [2007-2011]

BS, Computer Science

**Now Kennesaw State University*

04 / SELECTED RESEARCH

[1] **Promptions / Dynamic Prompt Middleware**. Designed and evaluated dynamic prompt-refinement controls that help users steer generative AI through contextual UI elements (n=16); released by Microsoft as the open-source Promptions framework. (CHIWORK 2025)

[2] **Provocations for AI-Assisted Knowledge Work**. Designed and evaluated an LLM-based intervention that generated critiques and alternatives during decision-making tasks, supporting critical and metacognitive thinking (n=24). (2025 preprint)

[3] **Wrex**. Designed and evaluated a programming-by-example interface that generated readable Python for data scientists in Jupyter notebooks (n=12), improving data-wrangling effectiveness. (CHI 2020, Best Paper Award)

See my CV for a full list of publications.